

HYDER MOHYUDDIN

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PORTFOLIO hydermohyuddin.com — 15+ projects with full write-ups and live demos, updated regularly and preferred over this resume for the fullest view of my work.

EDUCATION

Bachelor of Science, Major: Computer Science, Minor: Visual Arts, University of Chicago

Mar 2025

Coursework: Data Visualization, Theory of Algorithms, Discrete Math (Matrix Math), Systems Programming II, Complexity Theory, Data Science I (Pandas and Statistics), Art & Machine Intelligence, Advanced Conceptual Drawing, Actuated User Interfaces and Technology, Inventing, Engineering and Understanding Interactive Devices

COMPUTER SKILLS

Programming Languages: Python, Java, JavaScript, Assembly, C, C#, React, Complex OOP

Web Development: HTML, CSS, JavaScript, jQuery, Website Builders Figma and Squarespace

3D Modeling & Prototyping: Autodesk Fusion, 3D printing, hardware integration with Arduino & electronics

AI & Machine Learning: Proficient in PyTorch and Stable Diffusion for generative AI and model optimization

WORK EXPERIENCE

AI Software Engineer — Stable Diffusion Research, University of Chicago, IL

Dec 2023 – May 2025

Software Engineering & Optimization

- Enhanced system scalability, privacy, and security by architecting and leading the development of a Python server with secure authentication, and protecting against reverse engineering through code obfuscation techniques.
- Explored new possibilities in diffusion model generation by designing and building an out-facing web interface, handling all aspects of software development, including back-end, front-end, model control, and quality assurance.
- Achieved efficient string matching and trie-based compression by developing optimization algorithms, and enabled the exploration of randomized algebraic expressions in machine learning model transformations.
- Enhanced Stable Diffusion software with custom memory management, enabling real-time tensor adjustments and improving processing efficiency.
- Developed systems to improve data compression, storage efficiency, and model training through optimized file structuring with JSON and YAML.

Model Manipulation & AI Research

- Developed research tools for tensor modification, including random formula generation to create diverse image outputs and used AI to filter results based on similarity and novelty.
- Created an intuitive system for targeting specific model layers, enabling controlled modifications to fine-tune generative outputs.
- Engineered tools for interpolation and precise modification of small elements in generated images, allowing systematic experimentation with variations.
- Designed a method to layer Voronoi patterns across multiple generations, enabling ultra-high-resolution image creation beyond standard software limits.
- Built logging and analysis systems to track weight and bias factors, measuring average modifications, standard deviations, and trends to extract meaningful insights from model behavior.

Interface & Collaboration

- Implemented SPIM (Salavon's Pathology Inducing Machine), an interactive React application presenting the research on Professor Salavon's public website — **used to demonstrate the work to investors, helping secure funding** — and developed Python GUIs to test render parameters.
- Created unique visual effects by developing media manipulation tools using randomized filters and Voronoi tessellations.
- Managed effective web styling by delegating tasks to a junior designer (a Computer Science master's student at the University of Chicago) during a two-month collaboration.

Software Developer (Contract)

2024 – Present

- Revived Lavoët, a dormant Parisian trunk-maker founded 1869, as a working digital brand: an interactive catalogue-raisonné rendering every trunk as inline SVG, a parametric pattern studio that designs and exports the house canvas, and the historic

monogram rebuilt from a c.1930 photograph using **computer vision** — delivered as zero-dependency single-file applications.

- Built an **aerospace part-sourcing tool**: concurrent async scraping (Playwright + static fallbacks) across supplier sites with deduplication and timeouts, and a local llama.cpp model returning validated structured JSON.
- Developed **AutoLens**, a vehicle-identification app (Expo/React Native) — on-device image pre/post-processing with server-side model inference.

Web Developer (Contract)

2024 – Present

- Built and run the **Compassion Course** platform (React + Firebase) serving **50,000+ participants across 120+ countries** — scheduled lessons, a custom CMS the client's team publishes with, and a team dashboard with Kanban boards and shared whiteboards.
- Designed and built production sites for clients including two opposite-demographic brand landing sites — **Glo Pop** and **Gorilla Greens** — plus an **art-therapy practice**, an **interfaith charity initiative** with a nomination/registration pipeline, and a **poetry collection** — largely framework-free vanilla HTML/CSS/JS.

Software Project Lead, University of Chicago Booth School of Business, Chicago, IL

Dec 2023 – Nov 2024

- Streamlined transformation tasks by automating the processing of compiled SQL data using Python.
- Enabled interactive visualization of environmental data on a dynamic globe using JavaScript and Three.js.
- Improved statistical analysis efficiency by developing features to partition, average, and divide data clusters.
- Enabled seamless visualization of 200GB of environmental data on the globe using Three.js by implementing dynamic data visualization with different Leaflet palettes.
- Drove team alignment on next steps by leading weekly progress updates and summarizing key advancements.

Tutor, Chicago, IL

2016 – Present

- Self-employed Tutor: provided 1,000+ hours of individualized instruction to 50+ middle-school, high-school, and college students in math, physics, chemistry, and essay-writing.
- Code Platoon (NGO): Assisted in teaching coding bootcamp for evenings/weekends with 30 veterans and active duty service members.
- Code Your Dreams (NGO): Taught a 7-week introductory coding course to 40 inner-city high school students and trained inner-city high schoolers in Youth Initiated Mentoring.

Technology Support Assistant, University of Chicago Humanities Computing

2018 – 2021

- Imaged, secured, and maintained software for 100+ Humanities Division computers; automated OS and application deployment and provided IT support to faculty and staff.

SELECTED PROJECTS

- **Pocket Sous Chef** — iOS app published on the App Store: turns pantry photos into structured recipes via an AI text-and-vision pipeline (SwiftUI + SwiftData).
- **Typometry** — client-side typing-analytics web app: keyboard heatmaps, bigram timing, and behavioral profiling (React + Vite).

Full write-ups and live demos for all client work and projects above: hydermohyuddin.com/projects

VOLUNTEER EXPERIENCE

Founder, Justice Lecture Series Club, University of Chicago Laboratory Schools

2017 – 2018

- Organized 24 after-school sessions around Harvard's 'Justice with Michael Sandel' course and led discussions applying theories of justice to current legal and political controversies.

Squash Coach, MetroSquash (Chicago Non-profit)

2012 – 2016

- Weekly after-school squash training for inner-city students; proposed a student-run online squash shop teaching web design, accounting, marketing, and operations.

INTERESTS

Languages: English (fluent), Gujarati (fluent), Hindi (intermediate), French (intermediate), Arabic (beginner), Japanese (beginner – Kanji, Hiragana, Katakana)

Arts: Digital art, pencil sketching, stencil design, 3D design and printing, painting

Activities: squash, skiing, biking, chess, video games